

Schneider Joachim
Prof Scharf
IST363
Project 2 Written Summary

Link: <https://schneider-joachim.github.io/ist363/P2/index.html>

For this project, I built an astronomy-themed webpage that pulls live data from two different APIs. The first was the NASA API, which I used through two of its endpoints. One endpoint fetches the Astronomy Picture of the Day, giving me a daily space image or video along with its title, date, and a written explanation. The other endpoint connects to NASA's full image library, where users can search any space topic and get back a grid of photos with titles and descriptions. The second API was the Wikipedia REST API, which returns a structured summary of any Wikipedia article including the title, a paragraph of text, a thumbnail, and a link to the full page. Together these two sources gave the page a good mix of visual content and educational writing, all loaded in real time.

For handling async operations, every API call in the project is written as an async function that uses await to pause and wait for the server response before doing anything with the data. Each function also has a try and catch block around it just in case a request fails for any reason, the page would then show an error message instead of crashing. The APOD function is pretty important because it tries three different requests in a loop, and falls back to a backup date if the first attempt does not come back successfully, which keeps the page functional even when the API has issues.

In terms of structure, the page is divided into three sections. The first shows the daily NASA photo with its title, date, and explanation all pulled directly from the JSON response. The second section is a search tool that displays up to six image results in a grid, where each card is built from the title, photo, and a short preview of the description. The third section shows Wikipedia summaries based on whichever topic button the user clicks, updating the content dynamically each time without reloading the page.

The main challenge I ran into was the NASA APOD endpoint returning a server error, and this is where AI assistance helped a lot. The page kept failing to load the astronomy picture with a 500 error, and I wasn't sure whether the problem was in my code or coming from the API itself. I used Claude to help me figure out the issue, and it identified that the DEMO_KEY which is shared by all new users has a very low request limit that causes it to fail frequently when too many people are hitting the API at the same time.

Claude also suggested two fixes, registering for a personal API key through NASA's developer portal to get a higher request limit, and adding a retry loop in the APOD function so the page would automatically try alternate dates if the first request failed. Both of those solutions ended

up in the final code. Besides debugging, I also used Claude to help improve the overall structure and visual design of the site, including the responsive image grid layout and the Wikipedia section, which was not part of my original plan but made the project better. Overall, this was a fun project especially because I really enjoy Astronomy, and Artemis just launched not too long ago so I figured why not add to the astronomy hype train.